

AI Academic Recovery Agent

n8n Workflow Reference & Recovery Guide

Four published workflows - what each one does, how they connect, and how to pick the project back up from a cold start.

Item	Value
Live website	academic-recovery-agent.vercel.app
Repository	github.com/mohamadfayoumi1-cyber/academic-recovery-agent
n8n instance	mohamadfayoumi.app.n8n.cloud
Data store	Google Sheet "AI Academic Recovery System"
Workflows	4 published, all active
Document date	18 September 2026

How the system fits together

The browser never talks to Google or to Gemini. It only ever calls an n8n webhook. n8n owns every credential, every sheet write and every AI call, then hands the browser finished JSON to display.

```
Browser (static site on Vercel)
-> n8n webhook
    -> Google Sheets (all data)
    -> Gemini (analysis, extraction, summaries)
-> JSON back to the browser
```

Two consequences worth remembering: the frontend contains no academic arithmetic and no API keys, and any change to risk logic, grading or deadlines belongs in n8n, never in JavaScript.

The four workflows at a glance

Workflow	ID	Nodes	What it owns
project bootcamp (fixed)	GoD7hKR3XZ0BcgWV	61	Academic analysis, study planning, syllabus extraction, adding a course, grade entry with adaptive re-planning, and the HIGH/CRITICAL risk email.
2 - Auth (Signup + Login)	ypUGe4yrdbdOxuzw	10	Account creation and sign-in against the Students sheet.
3 - Chapter Summary	vNmENofDboWkBWkO	7	Turns an uploaded chapter PDF into a study summary.
4 - Course Admin	HBjI4v4Uq1lIHcoM	11	Editing an assessment deadline and permanently removing a course.

Every webhook the site calls

Path	Method	Workflow	Sent by
auth	POST	2 - Auth	Sign up and sign in
academic-analysis	POST	bootcamp	Dashboard, My Courses
syllabus-upload	POST	bootcamp	Add Course (PDF upload)
confirm-syllabus	POST	bootcamp	Syllabus review screen
update-grade	POST	bootcamp	Entering a grade
chapter-summary	POST	3 - Chapter Summary	AI Chapter Summarizer
course-admin	POST	4 - Course Admin	Edit deadline, remove course

All seven are POST with CORS Allowed Origins set to *. Opening one in the address bar returns 404 because that is a GET request; it does not mean the webhook is broken.

Two auth webhooks exist inside project bootcamp (fixed), on paths create-account and student-login. Nothing on the website calls them - the site has only ever used /webhook/auth in workflow 2. They are unused leftovers, not broken, and are safe to delete.

1. project bootcamp (fixed)

ID GoD7hKR3XZ0BcgWV 61 nodes Published

The largest workflow. Four independent pipelines share it, each starting at its own webhook.

Pipeline A - academic analysis (/webhook/academic-analysis)

```
Webhook -> Get student -> Get Courses -> Get Assessments
  -> Calculate Academic State (code: averages, weights, what is reachable)
  -> Academic Analyst Agent (Gemini: risk level, priority, reason)
  -> Planner Agent (Gemini: the weekly plan)
  -> Format Study Plan -> Save Study Plan
  -> Build Analysis Response -> Return Analysis
```

The arithmetic is done in code; the agents supply judgement and wording. Build Analysis Response merges the two and is where the response shape is decided.

Pipeline B - syllabus upload (/webhook/syllabus-upload)

```
Syllabus Upload Webhook -> Extract from File (PDF text, binary field 'file')
  -> Syllabus Extractor Agent (Gemini: course name, code, assessments)
  -> Validate Syllabus Extraction -> Return Syllabus Review
```

Validate Syllabus Extraction also renames the agent's assessment_name / deadline onto the name / due_date the website expects. Without that mapping the review screen shows every row blank and refuses to save.

Pipeline C - confirm the course (/webhook/confirm-syllabus)

```
Confirm Syllabus Webhook -> Prepare Confirmed Course -> Save Confirmed Course
  -> Prepare Assessments -> Append row in sheet -> Confirm Course Response
```

Pipeline D - grade entry and adaptive re-plan (/webhook/update-grade)

```
Grade Update Webhook -> Prepare Grade Update -> Update Assessment Grade
  -> re-read student, courses, assessments
  -> Recalculate Academic State
  -> Re-Analyze Academic State (Gemini)
  -> Adaptive Planner Agent (Gemini)
  -> Format + Save Adaptive Study Plan -> Grade Update Response
```

This is demo step 6. It was completely broken until three separate fixes were applied to its two agents - see 'The three-of-five pattern' later in this document.

Risk alert email branch

```
Academic Analyst Agent -> Filter Risk Courses -> Send Risk Alert Email
```

A second branch off the analyst, running alongside the planner. Filter Risk Courses keeps only HIGH and CRITICAL courses and attaches the student's name and email, which the Get student node already read at the start of the execution. Returning an empty list ends the branch, so nothing is sent when nothing is at risk.

Send Risk Alert Email is deliberately DISABLED. It has no Gmail credential attached. To switch it on: open the node, choose a Gmail credential, right-click the node and pick Enable, then Publish. Note there is no de-duplication - every Analyze click re-sends an alert for every HIGH or CRITICAL course.

2. Auth (Signup + Login)

ID ypUGe4yrdbdOxuzw 10 nodes Published

```
Webhook1 (path 'auth') -> Get Students -> Handle Auth -> New Account?
  yes -> Save New Account -> Respond to Webhook1
  no -> Respond to Webhook1
```

A single webhook serves both actions; Handle Auth switches on the request's "action" field, which is either "signup" or "login". Passwords are SHA-256 hashed in the browser before they are sent, and only the hash is stored.

The live trigger node is named Webhook1, not Webhook. A second, unrelated trigger called Webhook (path 'website') also exists in this workflow and feeds a small report chain. Code inside Handle Auth must reference \$('Webhook1'). Referencing \$('Webhook') silently breaks signup and login.

student_id is the student's own university ID, typed on the signup form. It is the key that joins Students to Courses to Assessments to StudyPlans. Handle Auth rejects a blank ID and rejects one that already exists.

3. Chapter Summary

ID vNmENofDboWkBWkO 7 nodes Published

```
Webhook (path 'chapter-summary')
  -> Extract Chapter Text (PDF text, binary field 'file')
  -> Chapter Summarizer Agent (Gemini + Structured Output Parser)
  -> Build Summary Response -> Respond to Webhook
```

Accepts multipart/form-data with fields file, student_id, course_id and course_name. Returns overview, key_concepts, important_definitions and key_points. The prompt forbids inventing content: if the chapter defines no terms it must return an empty list rather than make some up.

4. Course Admin

ID HBJl4v4Uq1lIHcoM 11 nodes Published

One webhook, two actions, chosen by the "action" field.

```
action = update_deadline
  Webhook -> Which Action? -> Update Deadline -> Deadline Response

action = delete_course
  Webhook -> Which Action? -> Read Courses -> Read Assessments
    -> Plan Deletions -> Delete Course Row -> Delete Response
    -> Split Assessment Rows -> Delete Assessment Rows
```

Deleting a course is permanent: the course row and every assessment row belonging to it are erased. Plan Deletions refuses to act unless the course belongs to the student who asked.

The Google Sheets node deletes by row number, so Plan Deletions maps the Nth record of a full-tab read back to sheet row N+1, and returns assessment rows in descending order - deleting a row shifts everything below it up, so bottom-up keeps the remaining numbers correct. This assumes no blank rows in the middle of a tab. The response is deliberately sent before the assessment rows are deleted, so that a course with no assessments still gets a reply.

The Google Sheet

Every tab is written by n8n. The column names below are a contract - renaming one silently breaks whichever node auto-maps onto it.

Tab	Columns
Students	student_id, name, email, password_hash, university, major, semester, weekly_study_hours, target_gpa, created_at
Courses	course_id, student_id, course_name, credits, target_grade, syllabus_id
Assessments	assessment_id, course_id, assessment_name, weight, max_grade, grade, deadline, status
Syllabi, CourseKnowledge, StudentGoals, Analysis, StudyPlans	supporting tabs

The sheet is shared as "anyone with the link can edit", and its URL appears inside the workflow JSON in a public repository. Restrict it to named people before presenting.

Traps that have already cost real debugging time

#	Trap
1	Every webhook node needs Options -> Allowed Origins = * or the browser blocks it while curl still works.
2	Respond bodies need a leading = or {{ }} is returned as literal text.
3	Extract from File reads a binary field called 'data' by default. The uploads send 'file'.
4	Structured Output Parser must use "Define using JSON Schema". Pasting a schema into the "Generate From JSON Example" box makes Gemini fail with "Model output doesn't fit required format".
5	No union types in the schema. "type": ["string","null"] breaks Gemini.
6	Model choice matters more than the API key. Preview models rate-limit almost immediately; use gemini-3.1-flash-lite.
7	Retry On Fail belongs on all FIVE agents, not three.
8	A Sheets read placed after a multi-item node runs once per item and returns the whole tab each time. Use Execute Once.
9	Publishing one workflow can silently deactivate another. Check both before demoing.
10	To open a sub-node on the canvas, single-click it then press Enter. Double-click often hits the connection line behind it.
11	Production executions do not show on the canvas. Use the Executions tab.
12	A POST webhook opened in the address bar returns 404. That is a GET request.

#	Trap
13	n8n cloud has a draft/publish model, and this is the big one. Saving a workflow - including through the API - only writes a DRAFT. The previously published version keeps serving live webhook traffic. Edits look saved, the canvas shows them, and production still runs the old code. The tell is the top-right button: "Publish" with an orange dot means a draft is pending; greyed "Published" means it is not. If a fix seems to have no effect, check that button before debugging anything else.
14	Committing is not pushing, and pushing is not deployed. Finish with <code>git status -sb</code> and then fetch the deployed file.
15	The three-of-five pattern. Five AI agents exist: three on the main path (Analyst, Planner, Syllabus) and two on the adaptive re-plan path (Re-Analyze, Adaptive Planner). Three separate fixes were applied to the first three and never to the last two - the Gemini model and API key, the output parser mode, and Retry On Fail. All three are corrected now, but when you change anything on an agent, change it on all five and grep the JSON to confirm. Every time this was missed the casualty was demo step 6, and it failed with a different error each time.
16	An imported workflow's Google Sheets nodes will not publish until the sheet is referenced by gid rather than by name, AND each node has been opened once so the editor re-validates it. The symptom is "N nodes have issues" with no visible error inside the nodes.
17	The gviz CSV endpoint silently blanks values whose type does not match the column's inferred type, which looks exactly like data loss. Use <code>/export?format=csv&gid=<gid></code> instead.
18	The n8n JSON files in the repository drift from what is live. Export and diff before trusting one.

Design decisions that are deliberate

Decision	Why
A course with no graded assessment is NOT_ANALYZED, not MEDIUM or CRITICAL	The guard is <code>completed_weight > 0</code> , decided in code rather than by the agent. Before it, a brand new course reported "0%" because <code>Number(null)</code> is 0, which pushed the model toward CRITICAL.
Every agent prompt opens with today's date	None of them knew the date. A syllabus line like "Midterm - November 6" carries no year, so the extractor guessed, often landing in a year already past.
The risk email reads the student's address from the existing Get student node	A second Sheets read would trip trap 8.
The Gmail node has <code>onError = continueRegularOutput</code>	A mail failure must never break the analysis response the browser is waiting on.
<code>student_id</code> is the real university ID	It used to be a generated S001, computed as <code>rows.length + 1</code> , which produced duplicate IDs as soon as any row was deleted.

Picking the project up after a shutdown

Nothing lives on the laptop. The website runs on Vercel, the workflows run on n8n's servers, and the data is in Google Sheets. Closing the laptop stops nothing - the site stays up and the webhooks keep answering.

Step 1 - open the three places the project lives

What	Where	Sign in with
The website	academic-recovery-agent.vercel.app	your student account
The workflows	mohamadfayoumi.app.n8n.cloud	your n8n login
The data	Google Sheet "AI Academic Recovery System"	your Google account
The code	github.com/mohamadfayoumi1-cyber/academic-recovery-agent	GitHub
Deployments	vercel.com/dashboard	Vercel

Step 2 - get the code back on a machine

```
git clone https://github.com/mohamadfayoumi1-cyber/academic-recovery-agent
cd academic-recovery-agent
```

There is nothing to install and nothing to build. It is plain HTML, CSS and JavaScript. Opening index.html in a browser works, and Vercel serves the same files. Read HANDOFF.md first - it is the living status document and is more current than this PDF.

Step 3 - check the system is actually alive

Run this from any terminal. It asks the live analysis webhook for a real student and prints the answer.

```
curl -s -X POST \
  https://mohamadfayoumi.app.n8n.cloud/webhook/academic-analysis \
  -H "content-type: application/json" \
  -d '{"student_id":"S001","available_weekly_study_hours":15}'
```

A JSON reply with overall_status and a courses array means n8n, Google Sheets and Gemini are all working. Replace S001 with a real student ID from the Students tab.

Step 4 - if something looks broken

Symptom	Look here first
A fix seems to have no effect	Trap 13. The top-right button in n8n - orange "Publish" means your change is only a draft.
The site is missing a change you made	Trap 14. Did the commit push, and did Vercel actually build it? Check vercel.com/dashboard - a deployment stuck on "Initializing" blocks every one behind it and must be cancelled.
"Model output doesn't fit required format"	Traps 4 and 5 - the output parser's mode and schema. Check all five parsers, not three.
Grades or names not saving	The sheet column names. A Sheets node that auto-maps silently drops any field whose name does not match a column.
Everything times out or rate-limits	The n8n trial may have expired, or Gemini is rate-limiting. Check the Executions tab for the real error.

Step 5 - restoring a workflow from scratch

The four .LIVE.json files in the repository's n8n/ folder are exact exports of the published workflows. To restore one: in n8n create a blank workflow, then use the ... menu -> Import -> From File. Re-attach the Google Sheets and Gemini credentials, because exports carry credential names but never the secrets. Then Publish - importing alone does not make a workflow live.

Before presenting: attach the Gmail credential and enable Send Risk Alert Email; restrict the Google Sheet's sharing; delete any leftover test courses; and record a backup video. Do not click Analyze repeatedly - each click is two Gemini calls.